

ASSIGNMENT - 1

1.

```
test_dir — -zsh — 97x24

[siddarthayadavguntla@siddarthas-MacBook-Air Downloads % mkdir test_dir ]
[siddarthayadavguntla@siddarthas-MacBook-Air Downloads % cd test_dir ]
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % touch example.txt ]
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % ls ]
example.txt
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % mv example.txt renamed_example.txt ]
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % ls ]
renamed_example.txt
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % ]
```

- **mkdir** - Used to create a new directory or folder.
- **cd** - Used to navigate from one directory to another.
- **touch** - Used to create a new empty file.
- **ls** - Displays the list of files and folders present in a directory.
- **mv** - Used to move or rename a file or directory.

2.

```
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % cat /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:*:-2:-2:Unprivileged User:/var/empty:/usr/bin/false
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
```

```
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % tail /etc/passwd ]
_aonsensed:*:300:300:Always On Sense Daemon:/var/db/aonsensed:/usr/bin/false
_modelmanagerd:*:301:301:Model Manager:/var/db/modelmanagerd:/usr/bin/false
_reportsystemmemory:*:302:302:ReportSystemMemory:/var/empty:/usr/bin/false
_swtransparencyd:*:303:303:Software Transparency Services:/var/db/swtransparency
d:/usr/bin/false
_naturallanguaged:*:304:304:Natural Language Services:/var/db/com.apple.naturall
anguaged:/usr/bin/false
_spinandd:*:305:305:SPINAND Daemon:/var/empty:/usr/bin/false
_corespeechd:*:306:306:CoreSpeech Services:/var/empty:/usr/bin/false
_diagnosticservicesd:*:307:307:Diagnostic Services:/var/empty:/usr/bin/false
_mds_stores:*:308:308:Spotlight File Metadata Index Daemon:/var/empty:/usr/bin/f
alse
_oahd:*:441:441:OAH Daemon:/var/empty:/usr/bin/false
siddarthayadavguntla@siddarthas-MacBook-Air test_dir %
```

```
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % head /etc/passwd ]
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
siddarthayadavguntla@siddarthas-MacBook-Air test_dir %
```

- **Cat** - Cat is used to display content inside the file
- **Head** - head is used to display the first 5 lines of the file content
- **Tail** - tail is used to display the last 5 lines of the file content

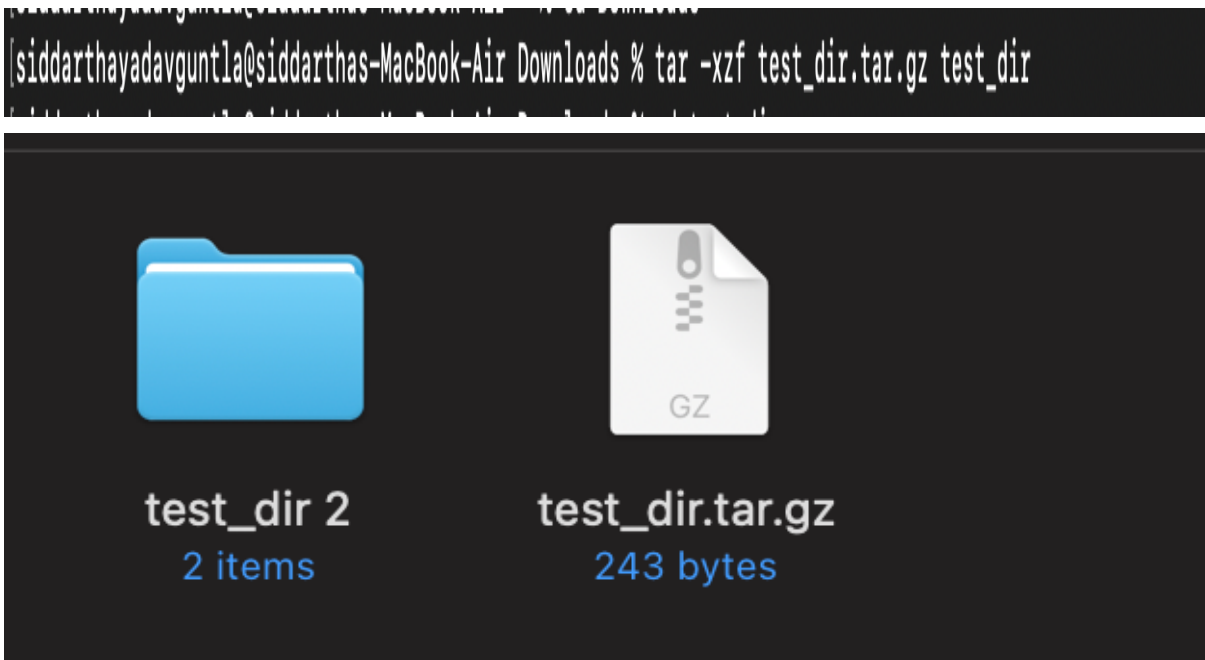
3.

```
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % grep "root" /etc/passwd ]
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
siddarthayadavguntla@siddarthas-MacBook-Air test_dir %
```

GREP - Global regular expression print is used to find a text pattern inside the file

4.

```
[siddarthayadavguntla@siddarthas-MacBook-Air test_dir % ls
renamed_example.txt      test_dir.tar.gz
siddarthayadavguntla@siddarthas-MacBook-Air test_dir %
```



- **Tar** - is used to combine many files and folders into one single file.

5.

```
siddarthayadavguntla@siddarthas-MacBook-Air test_dir % wget https://example.com/sample.txt
--2026-05-25 16:14:52-- https://example.com/sample.txt
Resolving example.com (example.com)... 2606:4700:9ae5:73da:acdd:db2:ef6b:ff98, 172.66.147.243, 104.20.23.154
Connecting to example.com (example.com)[2606:4700:9ae5:73da:acdd:db2:ef6b:ff98]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-25 16:14:54 ERROR 404: Not Found.
```

wget does

- Downloads files from the internet
- Saves them in the current directory

6.

```
siddarthayadavguntla@siddarthas-MacBook-Air test_dir % touch secure.txt
siddarthayadavguntla@siddarthas-MacBook-Air test_dir % chmod 444 secure.txt
siddarthayadavguntla@siddarthas-MacBook-Air test_dir % ls -l secure.txt
-r--r--r-- 1 siddarthayadavguntla staff 0 25 May 16:16 secure.txt
siddarthayadavguntla@siddarthas-MacBook-Air test_dir %
```

chmod is used to change who can read, write, or run a file in Linux.

7.

```
[siddarthayadavguntla@siddarthas-MacBook-Air ~ % export MY_VAR=Hello  
[siddarthayadavguntla@siddarthas-MacBook-Air ~ % echo $MY_VAR  
Hello  
siddarthayadavguntla@siddarthas-MacBook-Air ~ % █
```

- **export** is used to create an environment variable that can be accessed by programs and commands in Linux